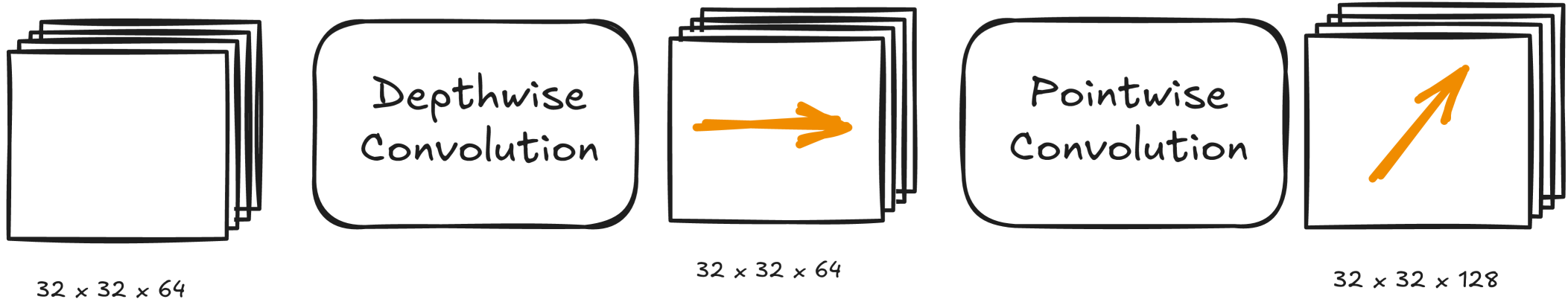


Depthwise Separable Convolution



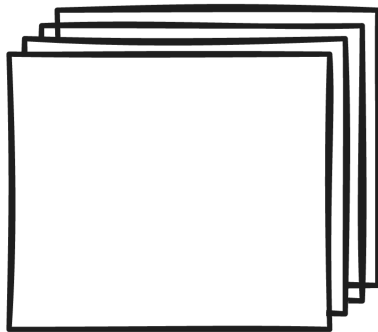
What, why & How with Code!

Depthwise Separable Convolution

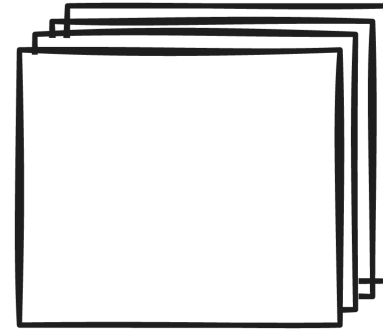
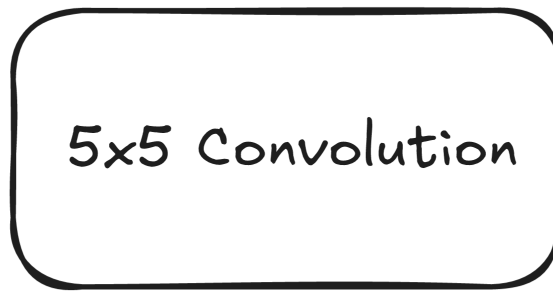
It is an efficient 2D convolution operation that significantly reduces the number of FLOPs and learnable parameters. This entails less processing power, less memory and faster inference at the slight cost of accuracy.

Depthwise Separable Convolution

It does so by splitting the convolution into 2 parts



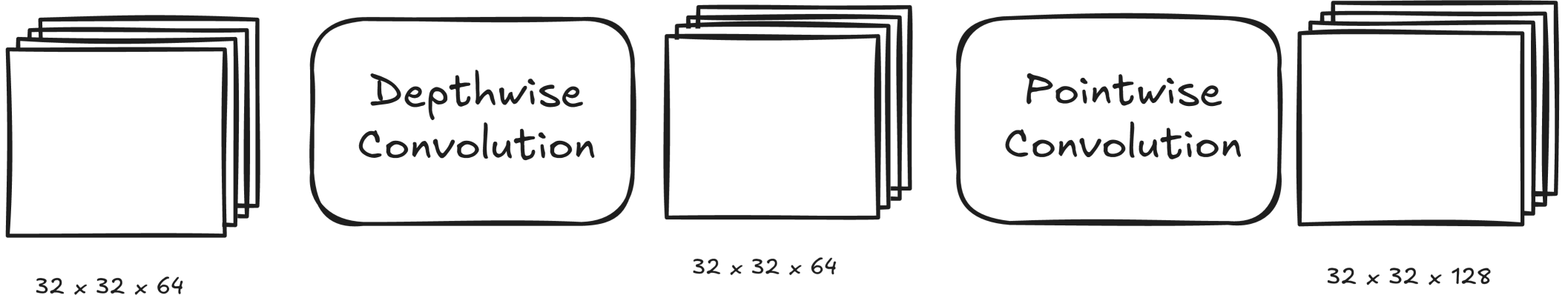
$32 \times 32 \times 64$



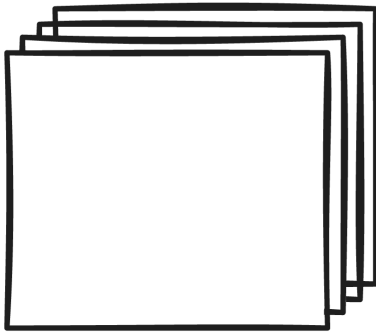
$32 \times 32 \times 128$

Depthwise Separable Convolution

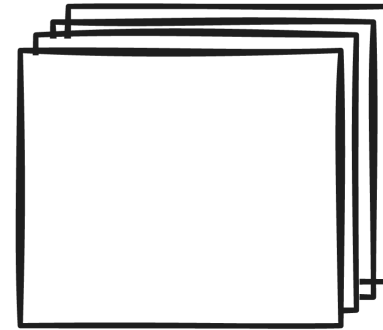
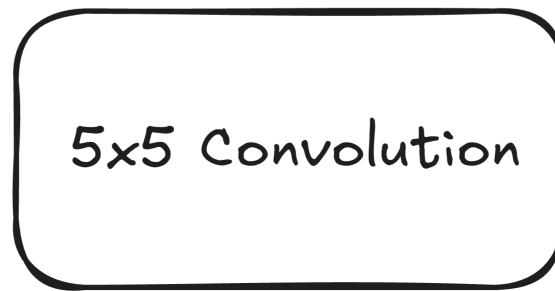
It does so by splitting the convolution into 2 parts



Convolution

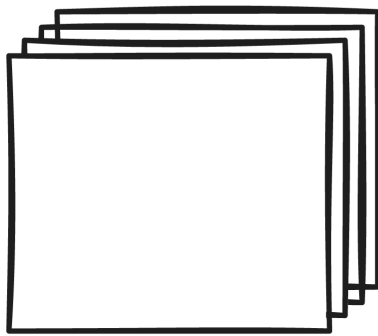


$32 \times 32 \times 64$

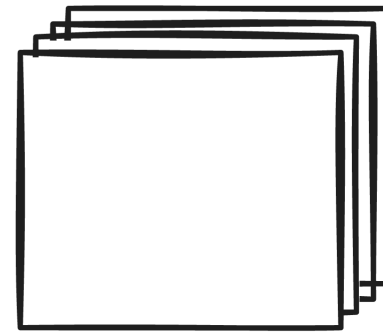
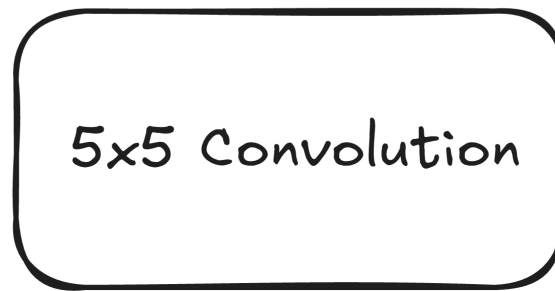


$32 \times 32 \times 128$

Convolution



$32 \times 32 \times 64$

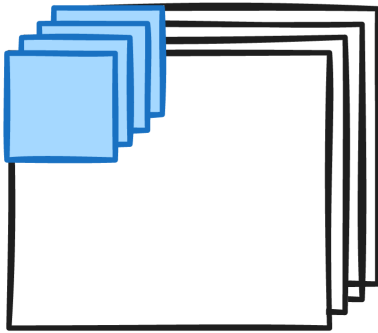


$32 \times 32 \times 128$

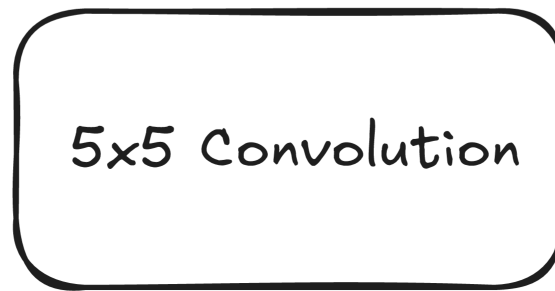
Let's see how to do with standard convolutions and then depthwise separable convolutions.

Full 2D Convolution

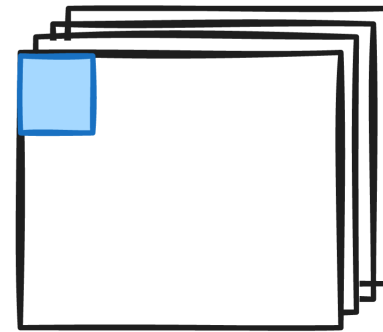
$5 \times 5 \times 64$ filter



$32 \times 32 \times 64$



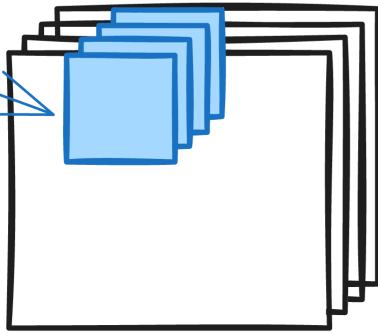
feature map output



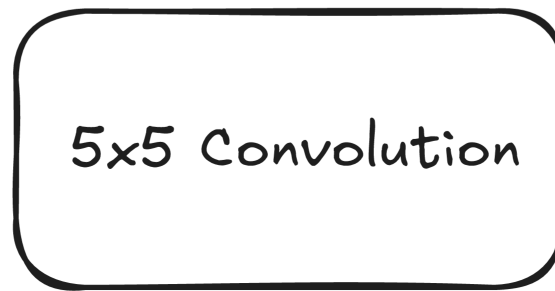
$32 \times 32 \times 128$

Full 2D Convolution

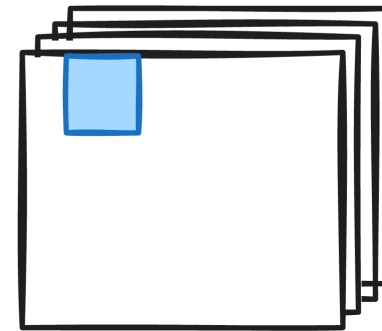
5 x 5 x 64 filter



32 x 32 x 64



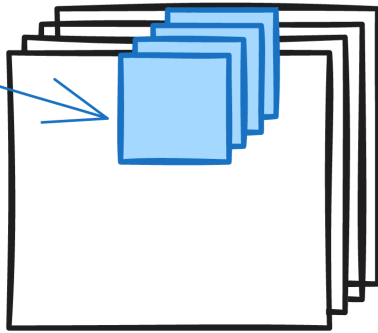
feature map output



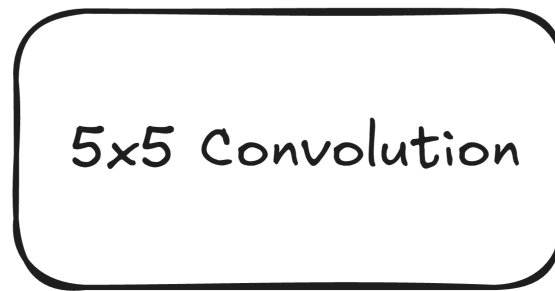
32 x 32 x 128

Full 2D Convolution

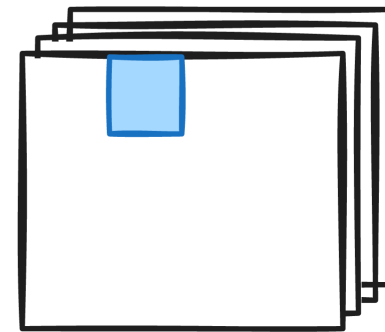
$5 \times 5 \times 64$ filter



$32 \times 32 \times 64$



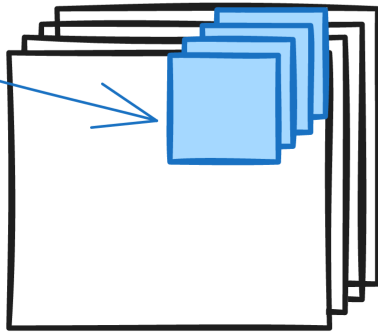
feature map output



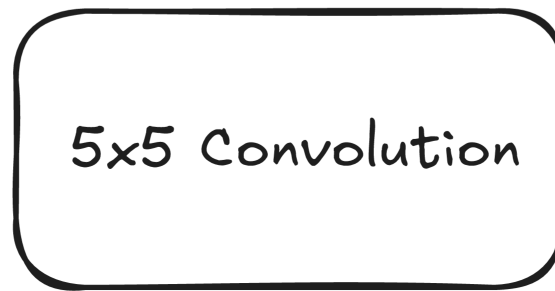
$32 \times 32 \times 128$

Full 2D Convolution

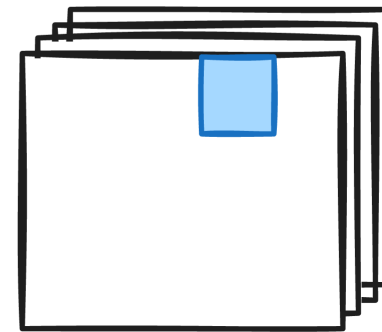
5 x 5 x 64 filter



32 x 32 x 64



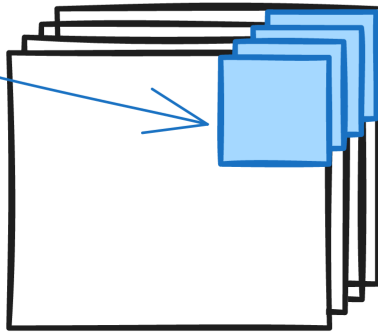
feature map output



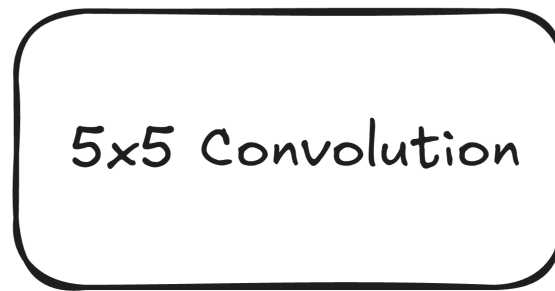
32 x 32 x 128

Full 2D Convolution

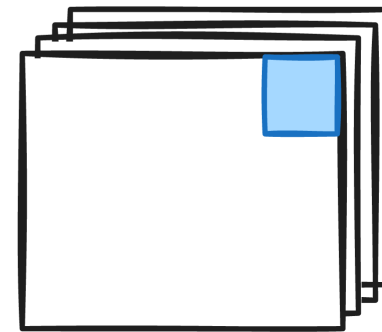
5 x 5 x 64 filter



32 x 32 x 64



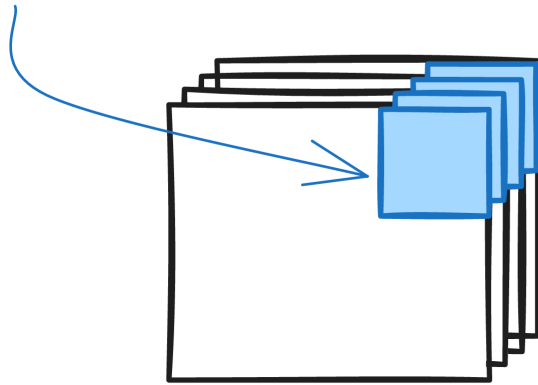
feature map output



32 x 32 x 128

Full 2D Convolution

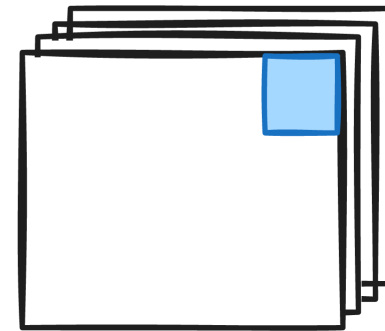
5 x 5 x 64 filter



32 x 32 x 64

5x5 Convolution

feature map output



32 x 32 x 128

Number of learnable parameters

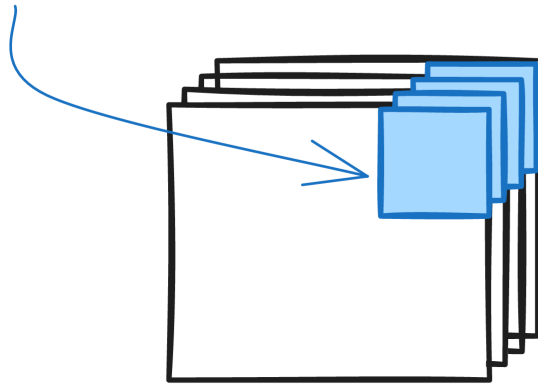
$$= 5 \times 5 \times 64 \times 128$$

$$= 204K$$

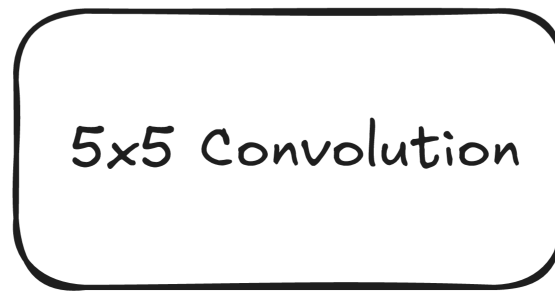
* bias terms excluded for simplicity.

Full 2D Convolution

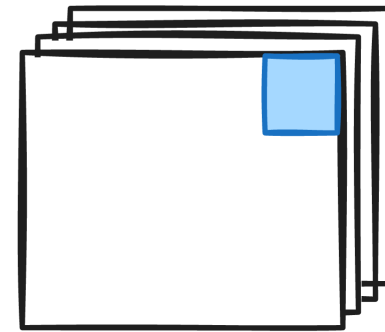
5 x 5 x 64 filter



32 x 32 x 64



feature map output



32 x 32 x 128

Number of learnable parameters

$$= 5 \times 5 \times 64 \times 128$$

$$= 204K$$

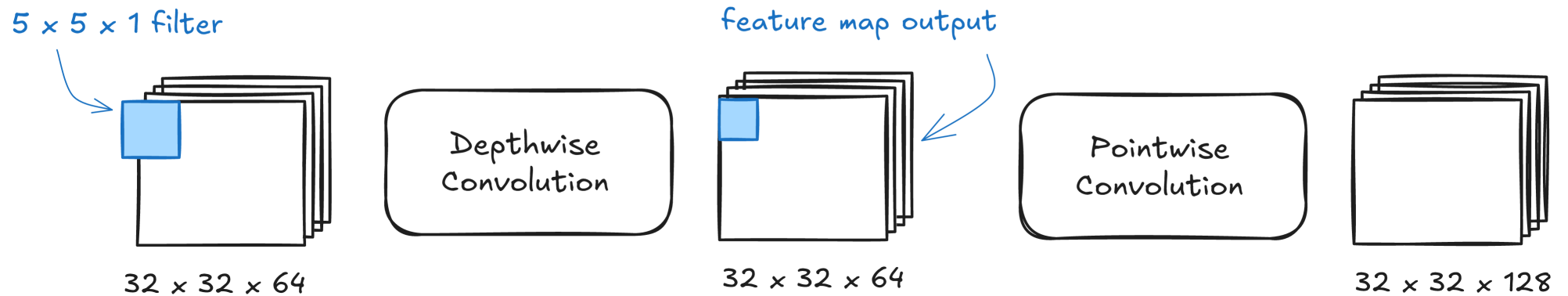
Number of multiplications

$$= 5 \times 5 \times 64 \times 32 \times 32 \times 128$$

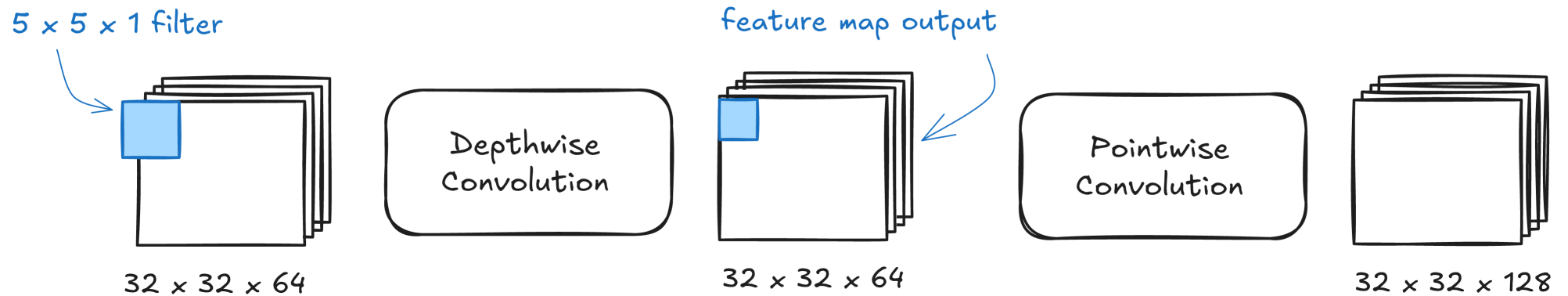
$$= 209M$$

* bias terms excluded for simplicity.

Depthwise Separable Convolution: How?

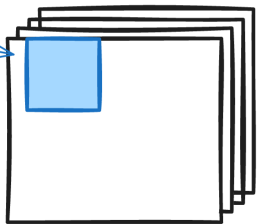


Depthwise Separable Convolution: How?



Depthwise Separable Convolution: How?

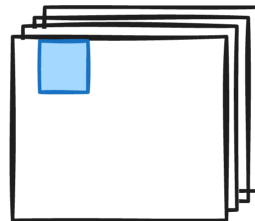
5 x 5 x 1 filter



32 x 32 x 64

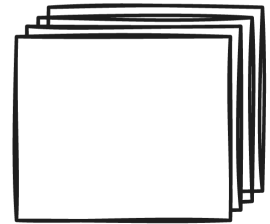
Depthwise
Convolution

feature map output



32 x 32 x 64

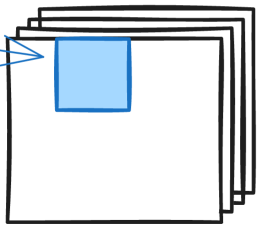
Pointwise
Convolution



32 x 32 x 128

Depthwise Separable Convolution: How?

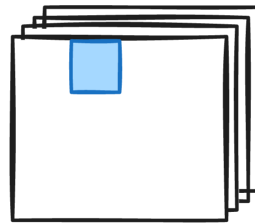
5 × 5 × 1 filter



32 × 32 × 64

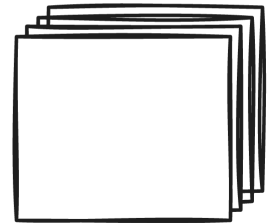
Depthwise
Convolution

feature map output



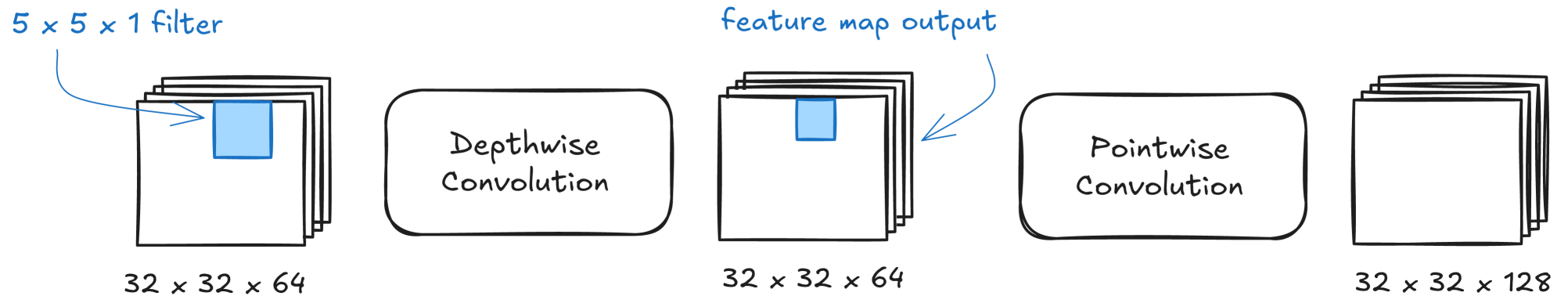
32 × 32 × 64

Pointwise
Convolution

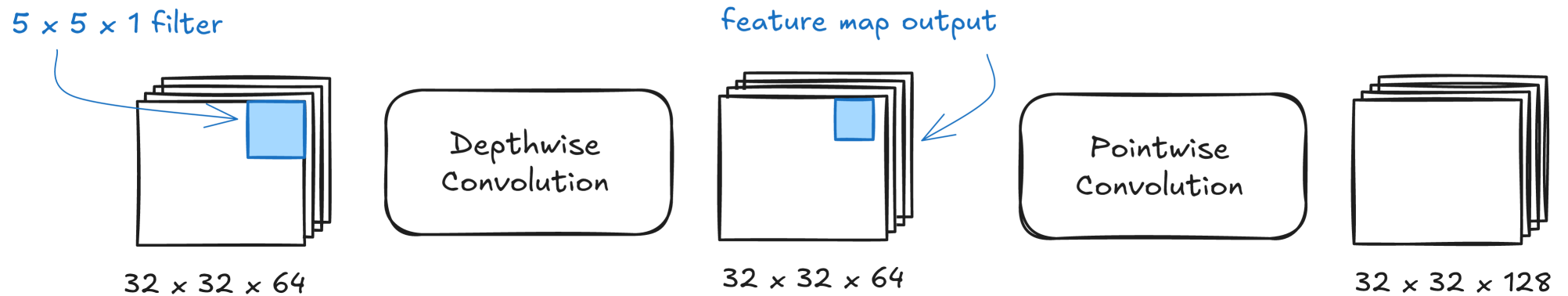


32 × 32 × 128

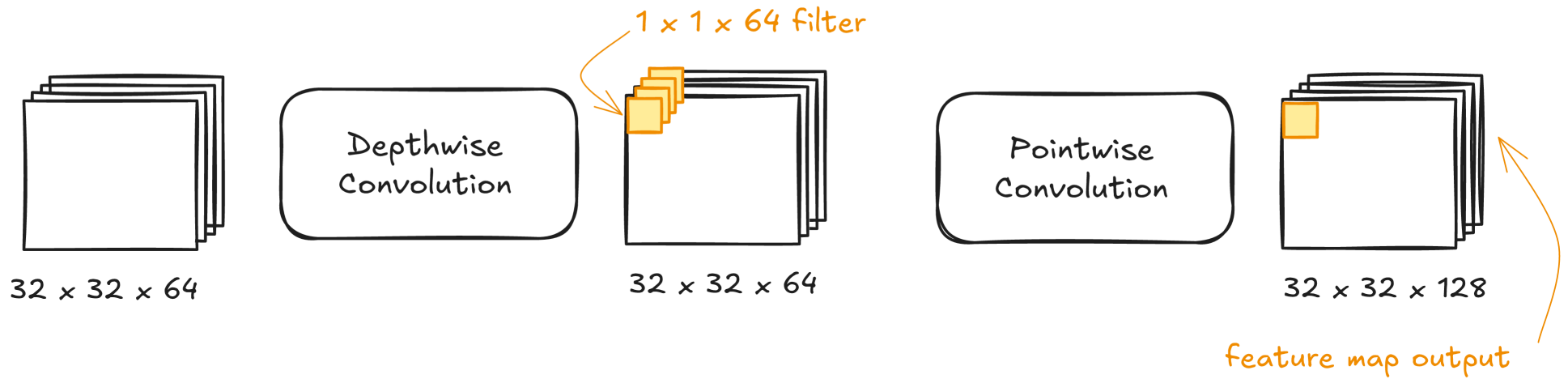
Depthwise Separable Convolution: How?



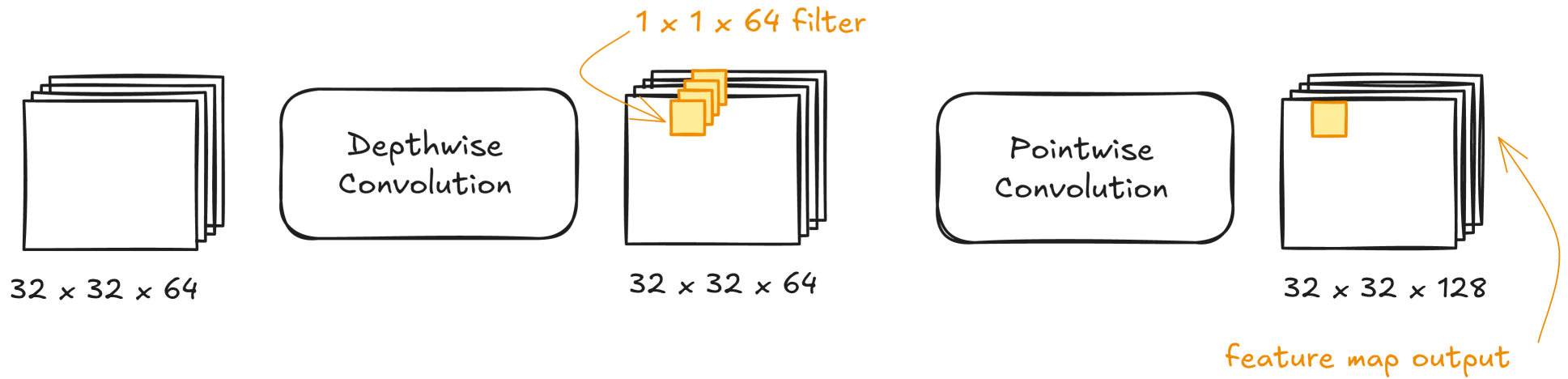
Depthwise Separable Convolution: How?



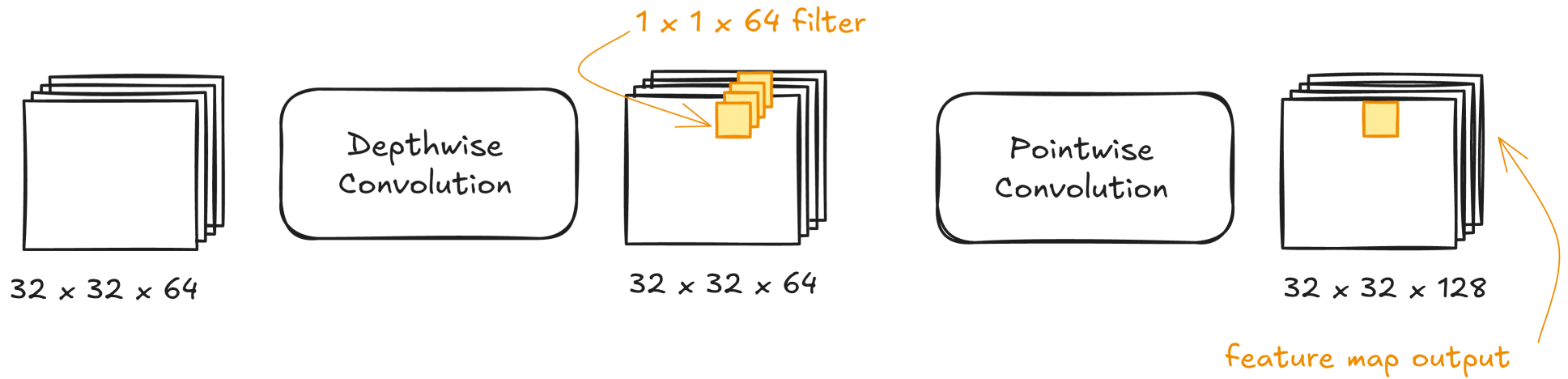
Depthwise Separable Convolution: How?



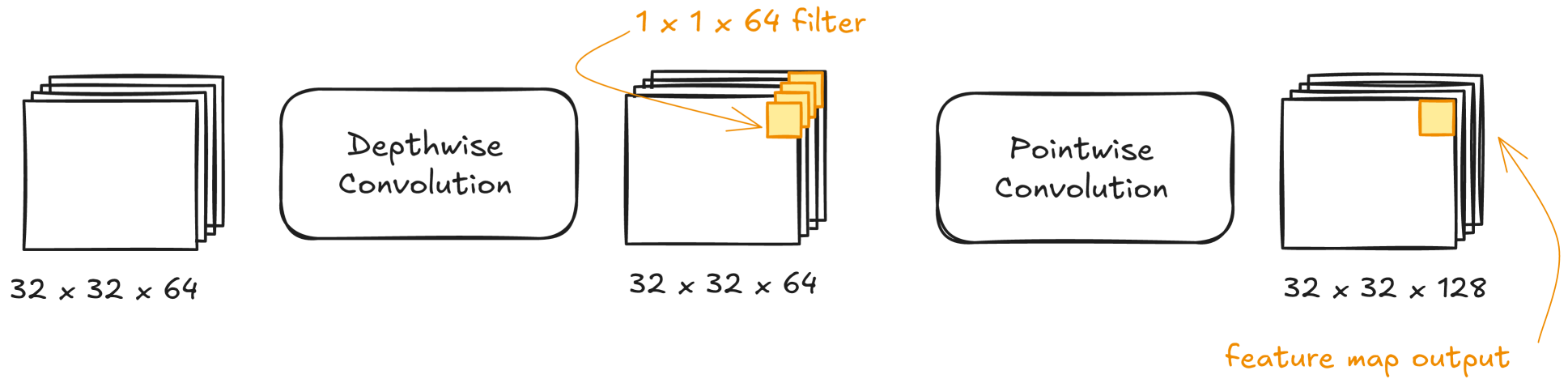
Depthwise Separable Convolution: How?



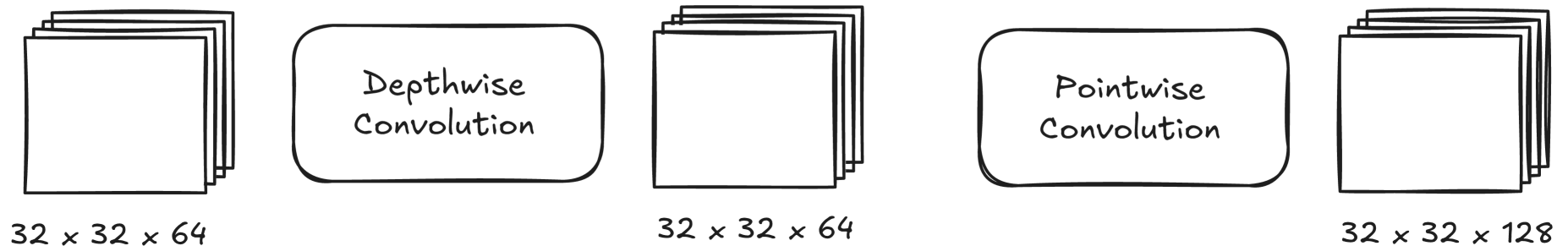
Depthwise Separable Convolution: How?



Depthwise Separable Convolution: How?



Depthwise Separable Convolution: How?



Number of learnable parameters

$$\begin{aligned} &= 5 \times 5 \times 1 \text{ (filter)} \times 64 \text{ (channels)} \\ &+ 1 \times 1 \times 64 \text{ (filter)} \times 128 \text{ (channels)} \end{aligned}$$

$$= 9,792$$

Number of multiplications

$$\begin{aligned} &= 5 \times 5 \times 1 \text{ (filter)} \times 64 \text{ (channels)} \times 32 \times 32 \text{ (windows)} \\ &+ 1 \times 1 \times 64 \text{ (filter)} \times 128 \text{ (channels)} \times 32 \times 32 \text{ (window)} \end{aligned}$$

$$= 10M$$

* bias terms excluded for simplicity.

Total Parameters and Comparison

Standard Convolution

$$K^2 \cdot C_{in} \cdot C_{out}$$

Depthwise Separable Convolution

$$K^2 \cdot C_{in} + C_{in} \cdot C_{out}$$

Reduction Ratio

$$\text{Reduction} = \frac{K^2 \cdot C_{in} + C_{in} \cdot C_{out}}{K^2 \cdot C_{in} \cdot C_{out}} \approx \frac{1}{C_{out}} + \frac{1}{K^2}$$

Use Cases

- Popular in mobile and efficient architectures:
 - **MobileNet (v1, v2)**
 - **EfficientNet**
 - **Xception**
- Helps reduce model size, memory, and inference time.
- Great for edge devices, real-time applications.

Depthwise Separable Convolution

Hence, we have this insight:

With DSC, we are able to significantly reduce the number of learnable parameters and multiplications

This entails less computation resources, less memory used and faster inference times.